A proof of the generalized power-sum identity

MathDB 334350, with an extension to every odd function

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Abstract

We prove Svinin's conjectured equality between a sum over weakly increasing tuples and a binomial combination of sums over ordered tuples. In fact, the equality holds when the odd power is replaced by any odd function on the integers. The proof establishes a Laurent polynomial identity by counting gaps and applying an elementary truncated negative-binomial identity.

Task 1: the original conjecture and its full solution

Let $n, j \geq 1$ be integers, and define

$$A(z) = z + \cdots + z^n, \qquad A(z)^\ell = \sum_{q=0}^{\ell(n-1)} C_{\ell,q}(n) z^{\ell+q}.$$

For $k \geq 0$, put

$$s_{k,\ell}(n) = \sum_{q=0}^{\ell(n-1)} C_{\ell,q}(n)(\ell+q)^{2k+1}, \qquad S_{k,j}(n) = \sum_{q=0}^{j-1} \binom{j(n+1)}{q} s_{k,j-q}(n).$$

Conjecture 3.1 of [1] asserts that

$$\sum_{1 \leq \lambda_1 \leq \cdots \leq \lambda_j \leq jn} \sum_{i=1}^{j} (\lambda_i - (i-1)n)^{2k+1} = S_{k,j}(n). \tag{1}$$

The source proves the case $j = 2$. We prove every j, without restrictions on k or n beyond those stated.

Theorem 1. *For every odd function $\psi : \mathbb{Z} \to \mathbb{C}$ and all $n, j \geq 1$,*

$$\sum_{1 \leq \lambda_1 \leq \cdots \leq \lambda_j \leq jn} \sum_{i=1}^{j} \psi(\lambda_i - (i-1)n) = \sum_{\ell=1}^{j} \binom{j(n+1)}{j-\ell} \sum_{r_1,\ldots,r_\ell=1}^{n} \psi(r_1 + \cdots + r_\ell). \tag{2}$$

In particular, (1) *holds for every $k \geq 0$.*

A finite coefficient identity

Lemma 1. *For positive integers a, j, define*

$$P(X) = \sum_{q=0}^{j-1} \binom{a+q-1}{q} X^{j-q}, \qquad T(b) = [t^{a-1}] \frac{1}{(1-t)^j (b-t)}.$$

As rational functions of b,

$$T(b) = b^{-a}(1-b)^{-j} - P\big((1-b)^{-1}\big). \tag{3}$$

Proof. The elementary polynomial identity

$$(1-b)^j \sum_{p=0}^{a-1} \binom{j+p-1}{p} b^p + b^a \sum_{q=0}^{j-1} \binom{a+q-1}{q} (1-b)^q = 1 \tag{4}$$

follows by partitioning binary words of length $a+j-1$ according to whether the jth second symbol or the ath first symbol occurs first. Give the first symbol weight b and the second weight $1-b$. In the first case, $p < a$ first symbols precede the last second symbol; in the second case, $q < j$ second symbols precede the last first symbol. The total weight of all possible suffixes is one. These two cases give the two sums, and the total weight of all words is $(b + (1-b))^{a+j-1} = 1$.

On the other hand, expanding at $t = 0$ gives $T(b) = b^{-a} \sum_{p=0}^{a-1} \binom{j+p-1}{p} b^p$. Divide (4) by $b^a(1-b)^j$ to obtain (3). □

The marked-tuple generating polynomial

Fix n, j, and abbreviate

$$a = jn, \qquad h = n+1, \qquad N = j(n+1) = a + j, \qquad B(z) = 1 + A(z).$$

Let

$$F(z) = \sum_{1 \le \lambda_1 \le \cdots \le \lambda_j \le a} \sum_{i=1}^{j} z^{\lambda_i - (i-1)n}, \qquad R(z) = \sum_{\ell=1}^{j} \binom{N}{j-\ell} A(z)^\ell.$$

Thus F is a Laurent polynomial and R is an ordinary polynomial with zero constant term. We will prove

$$F(z) - F(z^{-1}) = R(z) - R(z^{-1}). \tag{5}$$

The involution $\lambda_i \mapsto a + 1 - \lambda_{j+1-i}$ sends each exponent to h minus the corresponding reversed exponent. Therefore

$$F(z^{-1}) = z^{-h} F(z). \tag{6}$$

For a tuple λ, write its nonnegative gaps as

$$g_0 = \lambda_1 - 1, \quad g_i = \lambda_{i+1} - \lambda_i \ (1 \le i < j), \quad g_j = a - \lambda_j.$$

They sum to $a-1$, and $\lambda_i - (i-1)n = 1 + g_0 + \cdots + g_{i-1} - (i-1)n$. Summing each gap independently gives

$$F(z) = z[t^{a-1}] \sum_{l=0}^{j-1} \frac{z^{-nl}}{(1-zt)^{l+1}(1-t)^{j-l}} \tag{7}$$

$$= z[t^{a-1}] \frac{(1-t)^{-j} - z^{-a}(1-zt)^{-j}}{(1-zt) - z^{-n}(1-t)}. \tag{8}$$

The second equality is a finite geometric sum.

Work temporarily in the field $\mathbb{Q}(z)$ and put

$$b = \frac{z^n - 1}{z^{n+1} - 1}.$$

Multiplying (8) by $1 - z^{-h}$ and using (6) yields

$$\begin{aligned} F(z) - F(z^{-1}) &= [t^{a-1}] \frac{(1-t)^{-j} - z^{-a}(1-zt)^{-j}}{b-t} \\ &= T(b) - T(zb). \end{aligned} \tag{9}$$

For the second term substitute $w = zt$ in the coefficient extraction; the factors z^{a-1}, z, and z^{-a} cancel.

Now

$$1 - b = z^n(1 - zb), \qquad (1 - zb)^{-1} = B(z), \qquad (1-b)^{-1} = z^{-n}B(z) = B(z^{-1}).$$

Since $a = jn$, we also have

$$b^{-a}(1-b)^{-j} = (zb)^{-a}(1-zb)^{-j}.$$

These terms cancel when Lemma 1 is applied to (9), leaving

$$F(z) - F(z^{-1}) = P(B(z)) - P(B(z^{-1})). \tag{10}$$

Identifying the binomial combination

As an identity of polynomials in B, with $A = B - 1$, we have

$$\begin{aligned} R + \binom{N}{j} &= [w^j] \frac{(1+w)^N}{1 - (B-1)w} \\ &= [w^j] \sum_{\ell \geq 0} B^\ell w^\ell (1+w)^{N-1-\ell} \\ &= \sum_{\ell=0}^{j} \binom{N-1-\ell}{j-\ell} B^\ell = P(B) + \binom{N-1}{j}. \end{aligned}$$

Thus $R = P(B) - \binom{N-1}{j-1}$. The constant disappears upon subtracting the expression with z^{-1}, so (10) proves (5). Although rational functions were used to establish it, both sides of (5) are Laurent polynomials. There are no exceptional values of n, j or numerical substitutions to exclude.

Proof of Theorem 1. Apply to (5) the linear map sending z^r to $\psi(r)$. Oddness gives $\psi(-r) = -\psi(r)$ and $\psi(0) = 0$, so each side becomes twice its evaluation on F or R, respectively. Dividing by two proves (2). Expanding $A(z)^\ell$ and choosing $\psi(r) = r^{2k+1}$ gives exactly the source coefficients and (1). □

As a further consequence, if c_r is the number of marked tuples with exponent r minus the number with exponent $-r$, then

$$c_r = [z^r]R(z) > 0 \qquad (1 \leq r \leq jn).$$

Indeed, choose $\ell = \lceil r/n \rceil \leq j$; since $\ell \leq r \leq \ell n$, at least one ordered ℓ-tuple from $\{1, \ldots, n\}$ has sum r. This proves the positivity consequence noted conditionally in [2].

Task 2: a direct cancellation on marked tuples

Construct an explicit sign-reversing pairing between marked tuples with negative and positive exponents. Identify the unpaired positive objects with the ordered tuples and binomial choices on the right of (2). Such a construction would give a direct combinatorial explanation of the coefficient positivity proved above.

Task 3: the even part of the distribution

Determine a useful positive formula for $F(z) + F(z^{-1})$. Use it to express sums of even functions over the same weak tuples and to quantify the difference from the odd-function binomial transform. Include the mass at exponent zero.

Task 4: quantitative properties of the positive coefficients

Study the coefficient sequence $([z^r]R(z))_{r=1}^{jn}$. Determine its modes, whether it is unimodal or log-concave, and its limiting distribution when j or n grows. The strict positivity just proved is a starting point, not a solution to these further questions.

Task 5: a recurrence for the normalized power sums

Use the identity to investigate the source's separate Conjecture 5.1 on representing $S_{k,j}(n)$ by polynomials in j with coefficients rational in $n(n+1)$. Seek a recurrence uniform in n, while distinguishing the known special case $n = 1$ and the later partial results cited in the literature audit.

Verification and scope. Exact tuple counts, independent convolutions, and rational checks supplement the all-parameter proof. Tasks 2–5 are proposals. Review is by the authoring agent; no independent review or proof assistant is claimed.

Campaign eligibility. This is a full mathematical proof with zero new campaign credit pending the literature audit. A related 2019 paper concerning the same sums was accessible only as an abstract and first-page preview. Its remaining contents must be checked before eligibility can be cleared; see the accompanying literature audit.

References

[1] A. K. Svinin, *Conjectures Involving a Generalization of the Sums of Powers of Integers*, Experimental Mathematics **27**(4), 437–443; published online 5 April 2017, Conjecture 3.1. `https://doi.org/10.1080/10586458.2017.1306815`. Preprint: arXiv:1610.05387v2 (1 March 2017).

[2] A. K. Svinin and S. V. Svinina, *On Some Generalizations of Sum of Powers of Natural Numbers*, Russian Mathematics **63**(8) (2019), 25–37. `https://doi.org/10.3103/S1066369X19080048`. The Russian original, pages 37–38, equation (25), retains the target as a conjecture and states its conditional positivity consequence.